IU Transit & Navigation Hub: Product Requirements Document

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Product Target - IU Mobile iOS & Android Applications

Target Launch Window - December 2026

Status - Draft v2.0 for build planning

1. Overview

The IU Transit & Navigation Hub gives IU Bloomington riders a transit screen they can trust, so they stop leaving the IU Mobile app for ETA Spot, Google Maps or a rideshare.

It covers the IU Campus Bus and Bloomington Transit inside the existing IU Mobile iOS and Android apps. The product bet is simple: riders leave because they cannot trust the arrival time and cannot tell which stop serves their building. Fix those two problems first, and the cleaner map and faster filters make the app worth staying in.

- **What ships:** seven features, from building-to-stop search and traffic-adjusted ETAs to labeled stops, bus clustering and one-tap route chips (section 5).
- **How it ships:** an MVP of the P0 features plus the cheapest P1 features by December 2026, then the rest in a follow-up release (section 9).
- **Who reads this:** the product owner and the engineer building it. Each requirement has a testable done-when, and open questions are listed at the end (section 11).

2. Problem Statement and Evidence

Riders cannot trust the transit feature in IU Mobile, so they leave the app to get to class. Students and faculty at Indiana University Bloomington depend on campus and city transit across a large, spread-out campus, and the current in-app experience pushes them to external alternatives.

Core User Pain Points

Pain point	What the rider sees	Effect
Inaccurate and lagging ETAs	A bus showing "5 minutes away" for over 15 minutes, caused by delayed GPS sync and no live traffic accounting	Riders miss buses or wait when they should walk
"Ghost buses" and unclear fleet capacity	Buses vanish from the map or never arrive; no way to see how many buses are running a route	Riders cannot judge whether a bus is really coming
Missing campus spatial context	Stops shown as plain, unlabelled black dots, with no link to buildings such as Hodge Hall or Wells Library	Riders open Google Maps to find the right stop
Cluttered map and deep menus	Bus icons stacked on top of each other; route filtering buried in multi-step drawer menus	Active routes are hidden and filtering takes too many taps

Business & Product Impact: Because of these failures, students open ETA Spot in a web browser, switch to third-party navigation apps, or pay for rideshares (Uber/Lyft) to avoid being late. The result is high user churn and low trust in the core IU Mobile ecosystem.

What riders can use today

The data and the competing tools already exist, which shapes what IU Mobile should offer that they do not.

Option	What it offers	Source
ETA Spot (IU Campus Bus)	Real-time bus tracking for the B, E, F, W and X routes, with bus locations, distance to the next stop and service alerts, on iOS, Android and web. Launched August 2024.	IU Today, Oct 10, 2024

Option	What it offers	Source
Spot (Bloomington Transit)	Replaced Double Map after it ended support. Bloomington Transit said a public GTFS Real Time feed would let riders track buses in apps such as Google Maps.	WFIU/IPM, Feb 7, 2024
Link Transit (third-party iOS app)	Live map, ETAs, route filtering, favorites, arrival alerts and directions for Bloomington Transit, built on its public data feeds	App Store listing

The gap IU Mobile can own is one place for both systems, with building-level context and traffic-adjusted ETAs. Whether IU Campus Bus data is available as a feed IU Mobile can use is not confirmed in these sources, and it is the first open question in section 11.

3. Goals, Non-Goals and Success Metrics

The hub succeeds if riders trust the arrival time and find the right stop for a building without leaving the app.

Goals

1. **Trustworthy arrival times:** ETAs that reflect real positions and live traffic, and never freeze.
2. **No ghost buses:** a bus that stops reporting disappears from the map within 3 minutes.
3. **Building-first navigation:** a rider can start from a campus building, not a route letter.
4. **A readable map:** labeled stops, clustered buses and one-tap route filters.

Non-Goals (this release)

- Fares, payments or ticketing.
- Multi-leg trip planning that mixes buses with scooters, rideshare or driving.
- Push notifications and saved-stop alerts (candidate for the next release).
- Other IU campuses; this release covers IU Bloomington only.
- Predicting bus crowding; capacity is shown only where the data already provides it.

Success Metrics

Baselines are not public, so each one is measured before launch. Targets marked *proposed* are starting hypotheses to confirm with the engineer and IU.

Metric	Definition	Baseline	Target
ETA accuracy	Share of predicted arrivals within 1 minute of the actual arrival, measured at 5 minutes before arrival	Measure from current feed logs	80% (proposed)
Ghost-bus rate	Buses on the map with no GPS ping for more than 3 minutes	Measure from current feed logs	0
Location refresh	Time between bus position updates on screen	Not measured	Every 5–10 seconds
Time to first ETA	Seconds from opening the hub to seeing an ETA for a nearby stop	Not measured	10 seconds or less (proposed)
Building search success	Share of building searches that end with a stop selected	New feature	85% (proposed)
Weekly hub users	Unique IU Mobile users who open the transit hub in a week	Measure in current app	Increase over baseline; target set after baseline
Riders still using other apps	Share of surveyed riders who still open ETA Spot, Google Maps or a rideshare for the same trip	Survey before launch	Decrease over baseline; target set after baseline

4. Personas and Key Scenarios

Two riders drive the design: one who needs exact arrival times every day, and one who needs to know which bus goes where.

Primary Persona: The Daily Commuter

- **Demographics:** Undergraduate/Graduate student living off-campus or in distant dorms (e.g., McNut, Briscoe).
- **Core Goal:** Get from their starting location to back-to-back classes (e.g., Hodge Hall, Kelley School of Business) as quickly as possible without missing a bus.

- **Current Friction:** Frustrated by "ghost buses" showing 5-minute arrival times for 20 minutes, causing them to be late for exams or lectures.
- **Key Need:** Hyper-accurate ETAs, traffic delay adjustments, and active bus counts so they know whether to wait or walk.

Secondary Persona: The Casual Rider

- **Demographics:** Freshman, campus visitor, or student who usually walks/scooters but relies on buses during bad weather or long cross-campus trips.
- **Core Goal:** Quickly identify which bus route takes them directly to a specific building without having to memorize complex route letters.
- **Current Friction:** Confused by unlabelled black dots on the map and forced to switch to Google Maps to find out which stop is closest to their target building.
- **Key Need:** Direct building-to-stop search, persistent stop labels on the map layer, and simple route directions.

Key Scenarios

#	Persona	Situation	What the hub must do
1	Daily Commuter	Ten minutes before class, deciding whether to wait for a bus or walk	Open straight to the nearest stop with a live, traffic-adjusted ETA and the number of active buses on the route
2	Daily Commuter	A bus has shown "5 min" for a long time	Drop the bus from the map once its signal is lost for 3 minutes, so the rider sees the next real option
3	Casual Rider	Rain starts and the rider needs to reach Wells Library	Search the building, auto-pick the closest stop, and show the route and walking path
4	Casual Rider	In a basement lecture hall with weak signal	Keep the map, routes and stop schedules usable offline, and mark live data as unavailable instead of showing stale ETAs

5. Requirements and Acceptance Criteria

Seven requirements, three at P0 and four at P1. The "Done when" column is the test the engineer and I will use to call each one finished, and the "Release" column shows what is in the December MVP.

ID	Feature	Requirement	Done when	Priority	Release
FR-01	Building-to-Stop Search	Search by campus building name (e.g., "Wells Library"). Auto-select the closest bus stop and recommend the best route.	Every building in the campus dataset is searchable. Selecting one drops a pin, highlights the nearest stop, draws a dashed walking path from the stop to the building entrance, and lists routes serving that stop with live ETAs.	P0	MVP
FR-02	Traffic-Adjusted ETAs	Recalculate arrival predictions from live GTFS-RT vehicle positions combined with real-time Bloomington traffic congestion data.	ETAs refresh with every position update. A bus card shows any traffic delay in plain words (e.g., "+2 mins due to 10th St traffic"). If live data is lost, the ETA is hidden instead of frozen.	P0	MVP
FR-03	Accurate Fleet Counter	Show the number of buses currently operating on each route card.	The count equals the number of buses drawn on the map for that route, and only counts buses that pinged within the last 3 minutes.	P0	MVP
FR-04	Ghost Bus Purge	Flag and remove bus markers whose GPS signal drops or freezes for more than 3 minutes.	After 180 seconds without a ping, the icon fades out and leaves the map, the bus is dropped from ETAs and the count, and it returns on its next ping.	P1	MVP
FR-05	Persistent Map Labels	Show legible text labels on major stops without requiring a tap.	At standard zoom, high-traffic stops show their name (e.g., "10th & Fee"). Labels never overlap; lower-priority labels hide first.	P1	Release 2
FR-06	Bus Stacking Clustering	Group overlapping bus icons on the same route segment into one cluster badge.	Two or more overlapping buses show one badge with a count (e.g., "3x F"). Tapping it expands the individual buses.	P1	Release 2

ID	Feature	Requirement	Done when	Priority	Release
FR-07	One-Tap Route Chips	Horizontal, scrollable filter chips at the top of the screen for instant map filtering.	Chips for All, Route B, Route E, Route F and Bloomington Transit sit under the search bar. One tap filters the map and the route list to that route; All restores everything.	P1	MVP

Why Release 2 holds FR-05 and FR-06: both depend on tuning how the map lays out labels and badges across zoom levels, which takes the most testing and is the safest to ship after the data fixes (FR-01 to FR-04) are live.

6. UX/UI and Screen Specification

The hub is one screen with three layers: a search and filter bar on top, the map in the middle, and a draggable card at the bottom.

Screen Structure & Visual Layout

- **Top Navigation Bar (Sticky)**
  - **Search Input Field:** Full-width search bar displaying placeholder text *"Search campus building or stop (e.g., Hodge Hall, Wells Library)"*.
  - **Filter Chips Row:** A horizontally scrollable row of single-tap pill buttons directly below the search bar: All, Route B, Route E, Route F and Bloomington Transit. Selecting a pill filters the map to that route only.
- **Main Interactive Map View**
  - **Building Overlay Layer:** Light grey 2D/3D outlines of major IU campus buildings with legible name labels.
  - **Bus Stop Markers:** Distinct stop pins replace plain black dots. Major high-traffic stops show persistent text labels (e.g., *"10th & Fee"*) at standard zoom levels.
  - **Vehicle Location Badges:** Live bus icons in high-contrast route colors (e.g., Purple for Route F).
  - **Vehicle Cluster Badges:** When two or more buses overlap on a route segment, they merge into one circular badge showing the quantity (e.g., "3x F"). Tapping expands the cluster.
- **Bottom Draggable Sheet (Card Overlay)**
  - **Default / Collapsed View:** A "Nearest Stop" card based on the user's live GPS, showing the stop name, walking time, and live ETAs for incoming buses.
  - **Expanded View (Swiped Up):** A list of all active routes. Each route card shows:

- Route Letter & Name
- Total Active Buses currently running
- Chronological list of upcoming stops with nearby landmark tags (e.g., *"Stop: Fee Lane → Serves Kelley School"*)

Core User Interactions

- **Searching for a Building:** Typing "Hodge Hall" drops a target pin on Hodge Hall, highlights the nearest bus stop, and draws a dashed walking path from the stop to the building entrance.
- **Tapping a Bus Icon:** Opens a mini card with bus ID, speed/status, live capacity (if available), and real-time traffic delay (e.g., *"+2 mins due to 10th St traffic"*).
- **Handling Stale Data:** If a bus sends no GPS ping for more than 3 minutes, its icon fades out and leaves the map automatically.

Screen States

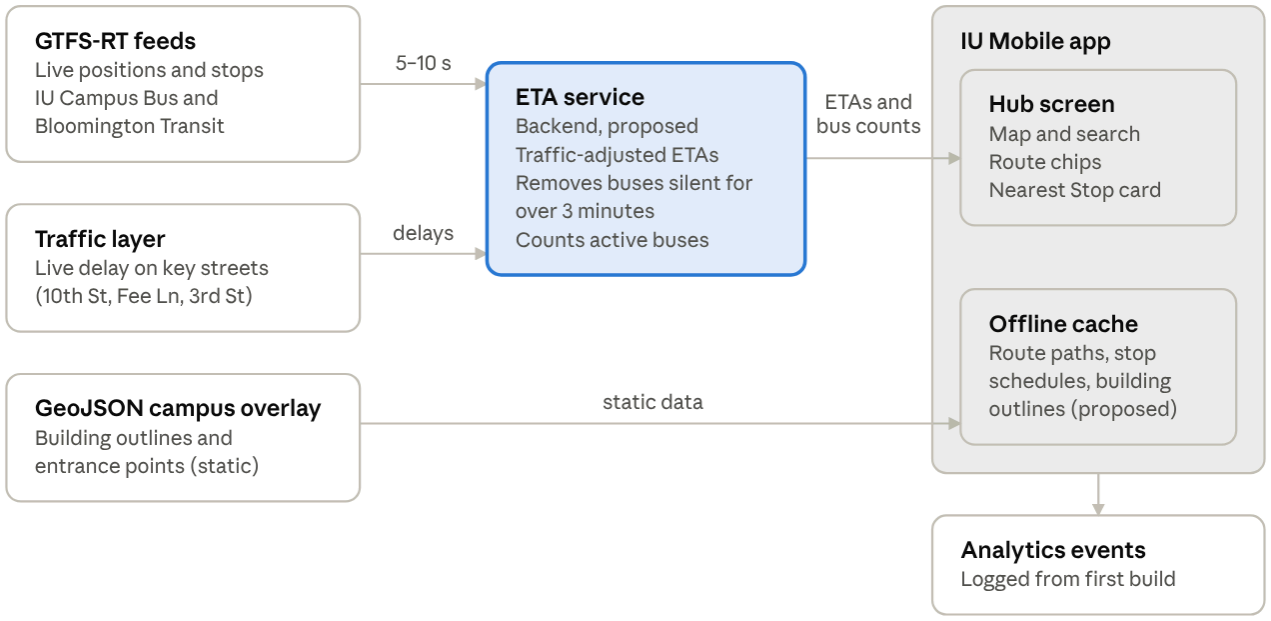
Every state below needs a design before build, because these are the moments a rider decides whether to trust the app.

State	When it happens	What the rider sees
Loading	Hub opens and live data has not arrived yet	Map and static stops show immediately from the cache; ETAs show a loading placeholder
Offline or weak signal	No connection or very slow connection	Routes and stop schedules from the cache, with a clear "Live data unavailable" banner and no ETAs
Stale feed	The whole feed stops updating, not just one bus	Banner with the time of the last update; ETAs hidden
No buses running	Late night, breaks or no service on a route	The route card says "Not running now" and shows the next scheduled service
Location off	The rider denied location access	Map centers on campus; the Nearest Stop card asks to enable location and offers search instead
No search result	The building name matches nothing	"No building found" with suggestions of close matches

7. Technical Architecture and Data Scope

Live bus data passes through one ETA service before it reaches the map, while static data stays on the phone so the app still works with a weak signal.

One ETA service turns live feeds into what riders see



data flow · 3 sources, 1 service, 1 app

Live feeds enter on the left, the ETA service adds traffic and removes silent buses, and the app shows the result. Whether that service runs on a small backend or on the phone is the engineer's call (section 11).

1. Data Sources & APIs

- **GTFS-Realtime (GTFS-RT) Feeds:** Primary data ingestion for live vehicle positions, route shapes, and stop sequences covering both IU Campus Bus and Bloomington Transit lines.
- **GeoJSON Campus Overlay:** Static geographic spatial dataset containing boundary outlines and entrance coordinate points for IU Bloomington buildings (e.g., Hodge Hall, Wells Library, IMU).
- **Traffic & Congestion Layer:** Integration with live traffic APIs (e.g., Google Maps Platform / Mapbox Traffic) to calculate dynamic delay vectors on key campus arteries (10th St, Fee Ln, 3rd St).

2. System Performance & Edge Handling

- **Location Ping Frequency:** Bus GPS locations update every 5–10 seconds to eliminate icon lagging across map tiles.
- **Data Invalidation Rules:** If a vehicle endpoint stops pinging for more than 180 seconds (3 minutes), the system marks the bus as "Offline" and removes it from the visible map interface to prevent "ghost bus" confusion.
- **Offline Caching & Fallback:** Local caching of static route paths and stop schedules ensures the app remains responsive during poor cellular reception in dense campus basements or lecture halls.

8. Analytics and Instrumentation

Every metric in section 3 needs an event behind it, so these are logged from the first build.

Event	Logged when	Key details	Feeds metric
hub_opened	Rider opens the transit hub	Entry point, location permission on or off, seconds until first ETA appears	Time to first ETA, weekly hub users
building_search	Rider submits a search	Search text, whether a result was chosen, seconds to choose	Building search success
stop_selected	Rider picks a stop	Source: search, map or Nearest Stop card	Building search success
eta_shown	An ETA is displayed	Route, stop, predicted minutes, traffic delay	ETA accuracy
arrival_observed	A bus actually reaches a stop (server side)	Route, stop, actual arrival time	ETA accuracy
bus_removed_stale	A bus is dropped for no signal	Route, bus ID, seconds since last ping	Ghost-bus rate
live_data_unavailable	Offline or stale-feed banner appears	Cause (offline or feed), duration	Reliability of live data
route_chip_selected	Rider taps a route chip	Route chosen	Feature use

Event	Logged when	Key details	Feeds metric
bus_tapped	Rider taps a bus icon	Route, whether capacity was shown	Feature use

ETA accuracy is the one metric that needs two events matched up: what the app predicted (eta_shown) and what happened (arrival_observed). Privacy rule: log no precise location tied to a named user, and confirm the final list with IU's privacy and data policies before launch.

10. Risks and Mitigations

The biggest risk is not the design but data access: if the live feeds are not available to us, the P0 features cannot work as written.

Risk	Why it matters	Mitigation
IU Campus Bus live data is not available as a feed IU Mobile can use	FR-02, FR-03 and FR-04 all depend on live vehicle positions	Confirm access with IU transportation staff and the ETA Spot vendor in week 1; if blocked, launch with Bloomington Transit data first
No access to the production IU Mobile app	A change inside the real app needs IU approval and its release process	Build the hub as a standalone prototype with the same screens and data, and hand over the spec if IU agrees to adopt it
Traffic data adds cost without improving ETAs	Traffic APIs are billed by usage, and campus routes are short	Fetch traffic only for the key arteries (10th St, Fee Ln, 3rd St), cache it, and compare traffic-adjusted ETAs with plain ones on the ETA accuracy metric before keeping it
A bus is removed during a short GPS gap	The 3-minute rule could hide a real bus that briefly lost signal	The bus returns on its next ping; log every removal so the threshold can be tuned
Wrong building entrance or stop data	A bad GeoJSON point sends a rider to the wrong stop	Test the stop recommendation on a sample of buildings before launch, and add a "Report a problem" link

Risk	Why it matters	Mitigation
Short timeline	About 11 weeks from now to December, with limited engineering time	MVP cut line in section 9: FR-05 and FR-06 move to Release 2 if the data work slips

11. Assumptions and Open Questions

Eight questions need an answer before or during week 1; the first two decide how much of the plan holds.

#	Question	Why it matters	Who answers
1	Does IU Campus Bus publish a GTFS-RT feed or API we can use?	Without it, FR-02, FR-03 and FR-04 cannot run on campus routes	IU transportation staff, ETA Spot vendor
2	Is this built into the production IU Mobile app, or as a standalone prototype?	Decides approvals, release process and how the engineer builds it	Ishan, IU contact
3	Which routes are in scope? This spec lists B, E, F and Bloomington Transit, but IU's tracker also covers W and X	Sets the route chips and the data to ingest	Ishan, IU
4	Which traffic provider and monthly budget?	Google Maps Platform and Mapbox are billed by usage	Ishan, engineer
5	Which stops count as "major" for persistent labels?	Needed for FR-05	IU ridership data, Ishan
6	Does a building and entrance dataset (GeoJSON) already exist, or do we build it?	Needed for FR-01; building it is extra work	IU GIS, engineer
7	Native iOS and Android, or one cross-platform codebase?	Affects the timeline and the map layer	Engineer

#	Question	Why it matters	Who answers
8	Does the feed include bus capacity or occupancy?	The bus mini card shows capacity only if available	Feed check by engineer

Working Assumptions

- Live positions arrive in the standard GTFS-RT format for both systems.
- Riders do not need to sign in to see transit information.
- Pre-launch baselines for ETA accuracy and ghost buses can be measured from feed logs.
- December 2026 means an MVP in riders' hands, not every feature.